NOVA SCHOOL OF SCIENCE & TECHNOLOGY

Interação Pessoa-Máquina

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Sous-Chef Ratchet: Kitchen Assistant

Stage 2: User and Task Analysis



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Problem Description

People who cook at home often have difficulty keeping track of the ingredients available in their kitchen, their quantities and expiration dates. This can make deciding what to cook somewhat difficult, lead to buying products that are already available at home, and contribute to food waste when ingredients are unused or even forgotten before they expire.

The cooking process itself also presents usability challenges. Users frequently need to consult recipes while their hands are occupied, wet or dirty, forcing them to interrupt the activity to interact with a mobile device. Furthermore, information about available ingredients, recipes and shopping needs is often managed separately, requiring users to switch between different tools or rely on memory

Users

The primary target users are university students and young adults who live independently or in shared households and cook at home regularly or occasionally. In particular, the project targets non-expert cooks who use mobile devices in their daily lives and would benefit from assistance in managing ingredients, choosing meals and following recipes.

User Classes:

1. Meal Discovery / Decision Users

These users use the app mainly to decide what they should cook, or discover new dishes to cook.

Main interests:

- Discover new recipes based on their inventory;
- Check for missing ingredients;
- Filter by available time;
- Filter by dish difficulty;
- Discover recipes that use up ingredients that are close to expiry;
- Compare alternatives.

Example:

> “I just got home, I have 30 minutes and I don’t want to go shopping. What can I cook?”

This class focuses on planning and execution. They are not necessarily trying to manage their kitchen inventory nor do they need help cooking. They are looking for help making a decision.

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2. Cooking Assistance Users

These users use the app mainly during the cooking process.

Main interests:

- Step by step instructions;
- Adapt portions;
- Consult quantities quickly;
- Timers;
- “Go back”/”repeat” steps buttons;
- Interactions with big buttons;
- Voice controls, hands free interactions.

Example:

> “I’ve chosen what I will cook. Now I just need help cooking the recipe, preferably without getting my screen dirty.”

This class would focus on the cooking aspect of the app.

Tasks

- **Get started on the app / Create an account** (Private or guest with easy access)

  <u>To Evaluate :</u>
  - Is it easy to start using the app ?
  - Are the account options simple and useful ?
  - Does it take an unnecessary amount of data to get started ?

- **Manage / Add owned products to the pantry** (Use the scan functionality or add the products manually. How do the users choose to complete this task? Does every option look simple or very complex ? How does the user prefer to do it?)

  <u>To Evaluate :</u>
  - Is it obvious and easy enough to add products ?
  - Are most options understandable / visible ?
  - Is this functionality practical ?

- **Choose a recipe and follow its execution** (Watch how the users search for recipes. Are recommendations good?)

  <u>To Evaluate :</u>
  - Is it hard to find a likeable option ?
  - Do options feel limited ?
  - Is the voice option accessible and useful ?

- **Create a shopping list**

  To Evaluate :
  - Is the option of adding ingredients to the list through cooking recipes accessible ? or is it preferable to add each ingredient one by one ?
  - Is the process practical and easy ?

Scenarios

1. **Creating an Account**:
   Bob wanted to create an account for the application. When entering the application, he was asked to login as a Guest or to add a new account. After pressing “Add a new account”, a register window appears. Bob had to insert a valid username and a password. Afterwards, Bob was sent back to the login screen, where he inserted the username and password he had created earlier. Bob was then allowed to view the application as a registered user.
2. **Adding Products to the Pantry:**
   Maria is a student who shares an apartment with two other people. She occasionally buys products unnecessarily or discovers that ingredients have expired before she uses them.
   Maria decides to organize the ingredients in her kitchen using the application. She opens the pantry section and attempts to add several products, including milk, eggs and rice. She notices that she can either scan a product or enter its information manually. She chooses the method she finds most convenient and follows the corresponding steps.
   While adding the products, Maria considers the information she needs to record, such as the quantity and expiration date. Some products may be easier to identify than others, and their expiration dates might not be available through scanning. She completes the entries and returns to her pantry to check whether the products have been registered correctly.
3. **Finding a Recipe with Available Ingredients:**
   Ana is hungry, has approximately 30 minutes to prepare dinner and does not want to go shopping. She also has some vegetables in her refrigerator that are approaching their expiration dates, so she would prefer to use them before they go to waste.
   Instead of browsing recipe websites individually or trying to remember what she can cook, Ana opens the application and looks for recipe suggestions based on her available ingredients. She checks the suggested dishes and considers whether they fit her available time, cooking skills and preferences.
   One suggestion looks promising, but Ana notices that she is missing an ingredient. She checks whether the ingredient is essential or whether she can prepare the dish with what she already has. She compares the available alternatives and chooses the recipe that best fits her needs.
   Before starting to cook, she checks the ingredients and instructions to make sure she has everything required.

4. **Following a recipe while cooking:**
   Daniel has selected a recipe through the application and is ready to prepare dinner. He is not an experienced cook, so he relies on clear instructions and frequently checks quantities and preparation steps.
   He places his phone near the kitchen counter and begins preparing the ingredients. As he cooks, his hands become wet or covered with food, making it inconvenient to touch the screen repeatedly, or to keep wiping his hands. He needs to check the next step, confirm the required quantity of an ingredient and set a timer for one of the cooking stages.
   Daniel attempts to navigate through the recipe without losing track of his progress. When he is unsure about an instruction, he wants to review the previous step. He also explores whether voice interaction is available and whether it allows him to access instructions without touching the device.
   After completing the recipe, Daniel finishes cooking without having to repeatedly interrupt the activity to search for information elsewhere.
5. **Creating a shopping list:**
   Sofia wants to prepare a dish over the weekend and has found a recipe that she likes in the application. Before going to the supermarket, she needs to determine which ingredients she must buy.
   She opens the recipe and reviews its ingredients. However, she does not want to purchase products she already has at home. She checks her pantry and compares the listed ingredients with the products currently available, including their quantities.
   After identifying the missing ingredients, Sofia adds them to her shopping list. She checks the list for duplicated products and confirms that it contains everything required for the recipe. At the supermarket, she consults the list while shopping so that she can purchase the necessary products without relying entirely on memory.
   This scenario investigates whether users can create shopping lists efficiently and whether combining recipe information with pantry information reduces manual work. It also examines whether users understand how to add, remove and verify items and whether the resulting list is sufficiently clear for use while shopping.

Interviews

Interview Questions and Results:

A. Background and Cooking Habits

1. **How often do you cook at home?**
   University students do not cook at home as often as older adults, and may have irregular routines. Adults cook at home much more often.
2. **Do you usually cook for yourself or for other people?**
   University students cook mainly for themselves but may also occasionally cook for roommates, while older adults cook more for their families, or occasionally for themselves.

B. Pantry and Ingredient Management

1. **How do you keep track of what ingredients you have at home?**
   Students tend to rely more on memory, but everyone seems to note down whatever ingredients they ran out of.
2. **How do you keep track of expiration dates?**
   Everyone checks a product's expiry date before using it, but may forget about them before needing them.
3. **What do you do before going shopping to determine which products you need?**
   Students go shopping more spontaneously and rely more on memory, older adults rely on shopping lists.

C. Recipe Discovery and Meal Planning

1. **How do you usually find recipes when you do not know what to cook?**
   Most people search online for recipes or stumble upon recipes while scrolling through social media. Older people rely more on recipe books or magazines.
2. **What do you do when you want to prepare a meal but are missing some ingredients?**
   Most people give up on cooking a meal when ingredients are missing. However, students seem more likely to immediately go shopping for those missing ingredients.
3. **How do you decide whether a recipe is suitable for the time you have available?**
   Most people consult the recipe's estimated cooking time, and prefer choosing recipes that do not take too long to cook. However, experienced cooks with more time on their hands are not very worried about cooking time.
4. **Would information about ingredients approaching their expiration dates influence your choice of meal? Why or why not?**
   Most people agreed that knowing exactly what ingredients are about to expire would encourage them to use them before the expiry date.

D. Assistance During Cooking

1. **How do you follow a recipe while preparing a meal?**
   Students are more likely to follow video guides, while others prefer step-by-step guides (online or physical books).
2. **What device or resource do you normally use to consult instructions?**
   Students depend entirely on their smartphones, while others may use magazines or cookbooks.
3. **How do you manage cooking times and timers?**
   Almost everyone relies on their smartphone if they have to set timers. Older adults may rely on clocks and memory, and sometimes even guess.
4. **Would you use voice interaction to consult or navigate a recipe while cooking? Under what circumstances would it be useful or inconvenient?**

People that used smartphones for recipes liked the voice controls idea for navigating recipes while cooking. Others were somewhat indifferent.

E. Shopping Lists

1. **How do you usually create a shopping list?**
   Most people add products to a shopping list as they run out of ingredients, and may add other products when they are interested in specific recipes.